

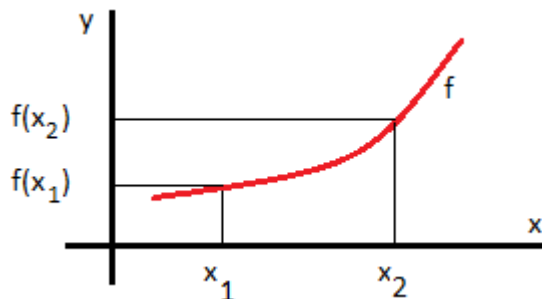
4.3 How the derivative affects the shape of the graph (Pages 293 – 304 in the text book)

PART I: Increasing functions, decreasing functions and the First Derivative Test for local extremes

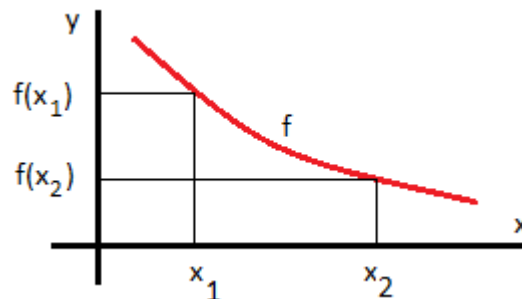
Let's first recall the definitions of an increasing and a decreasing function: (page 19 of the text book)

Definition: We say a function f is increasing on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I .

We say a function f is decreasing on an interval I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .



f is increasing



f is decreasing

Thus, a function is increasing over an interval if the function values increase over the entire interval and a function is decreasing over an interval if the function values decrease over the entire interval.

The derivative of a function assists us to determine where the function is increasing or decreasing: This is given by the Increasing / Decreasing Test, which you can find on page 293 of the text book.

Increasing / Decreasing Test

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval
- (b) If $f'(x) < 0$ on an interval, then f is decreasing in that interval

Proof of (b): Suppose $f'(x) < 0$ on an interval I . We must then show that f is decreasing on I .

Choose any two points x_1 and x_2 in I such that $x_1 < x_2$.

Since $f'(x) < 0$ on the interval I , we know that

1. f is differentiable on $[x_1, x_2]$ and so also on (x_1, x_2)
2. f is continuous on $[x_1, x_2]$ (Remember: f differentiable $\Rightarrow f$ continuous)

So, by the Mean Value Theorem, there exists a number c in (x_1, x_2) such that $f'(c) = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$

Thus, $f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$.

Since $f'(c) < 0$ (given) and $(x_2 - x_1) > 0$ (because $x_2 > x_1$), it follows that $f(x_2) - f(x_1) < 0$, or $f(x_1) > f(x_2)$.

This shows that f is decreasing.

Example 1: Find the interval(s) on which f is decreasing, if $f(x) = xe^x$.

Solution: Note that $f'(x) = e^x + xe^x < 0 \Rightarrow e^x(1 + x) < 0$

Since $e^x > 0$ for all $x \in \mathbb{R}$, $e^x(1 + x) < 0 \Rightarrow (1 + x) < 0 \Rightarrow x < -1$

Thus, f is decreasing on the interval $(-\infty, -1)$

Example 2 is for Homework: (Will be discussed in the online lecture)

Example 2: Find the interval(s) on which f is increasing, if $f(x) = e^{2x} - 6x$.

The first derivative of a function also assists us to find the local extremes of the function. This is formulated in the First Derivative Test, which you can find on page 294 of the text book.

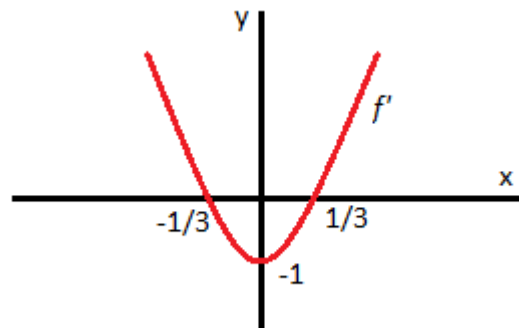
Example 3: Find the local extremes of the function f with $f(x) = 3x^3 - x$

Solution: We first need to find the critical numbers of f . Remember, the critical numbers are the points where either the derivative does not exist, or where the derivative is zero.

We note that $f'(x) = 9x^2 - 1 = 0 \Rightarrow x = \pm 1/3$.

Thus, the derivative of f exists for all $x \in \mathbb{R}$, and so the only two critical numbers are $x = 1/3$ and $x = -1/3$.

In order for us to use the First Derivative Test, we need to find the intervals on which f' is positive or negative. Let's use the graph of f' to do this:



As you can see from the graph: $f'(x) > 0$ if $x < -1/3$; $f'(x) < 0$ if $-1/3 < x < 1/3$ and $f'(x) > 0$ if $x > 1/3$.

Thus, since f' changes from positive to negative at $-1/3$, f has a local maximum at $-1/3$.

Also, since f' changes from negative to positive at $1/3$, f has a local minimum at $1/3$.

Example 4 is for Homework: (Will be discussed in the online lecture)

Example 4: Find the local extremes of the function f with $f(x) = -x^5 + 15x^3$